H11: Radiation

11.1 dipole radiation

11.1.1 what is radiation?

radiation

when charges accelerate their fields can transport energy irreversibly out to infinity

consider a source at the origin, radiating energy
> imagine a giant sphere of radius r around this source
> the power passing through its surface is the integral of the Poynting vector:

$$P(r,t) = \oint \mathbf{S} \cdot d\mathbf{a} = \frac{1}{\mu_0} \oint (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}.$$

however, these lightrays travel at the speed of light
> the em wave actually left the source an earlier time t_0=t-r/c, thus:

$$P_{\text{rad}}(t_0) = \lim_{r \to \infty} P\left(r, t_0 + \frac{r}{c}\right)$$

radiation of static sources

The area of a sphere is $4\pi r^2$
> for radiation to occur the Poynting vector must decrease no faster than $1/r^2$
> according to Coulomb, electrostatic fields fall of like $1/r^2$
Biot-Savart, magnetostatic fields fall of like $1/r^2$
> S falls of like $1/r^4$ for static configurations
> static sources don't radiate

>> for radiation we need to consider **E** and **B** fields that go like 1/r

11.1.2 electric dipole radiation

oscillating dipole

consider a dipole spinning around its axis at a angular frequency ω
> the charge is time dependent q(t):

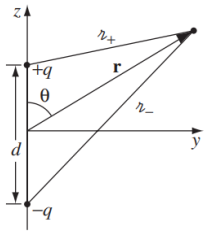
$$q(t) = q_0 \cos(\omega t).$$

the dipole moment is then, with maximum $p_0=q_0d$:

$$\mathbf{p}(t) = p_0 \cos(\omega t)\, \hat{\mathbf{z}},$$

the retarded potential is then:

$$V(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_0 \cos[\omega(t - \mathscr{r}_+/c)]}{\mathscr{r}_+} - \frac{q_0 \cos[\omega(t - \mathscr{r}_-/c)]}{\mathscr{r}_-} \right\}$$



and via the law of cosines:

$$\mathscr{r}_\pm = \sqrt{r^2 \mp rd\cos\theta + (d/2)^2}.$$

we can make approximations:
1: d<<r
2: d<<c/ω
3: r>>c/ω

1: d<<r

to the first order of d, we have:

$$\mathscr{r}_\pm \cong r\left(1 \mp \frac{d}{2r}\cos\theta\right)$$

from which follows:

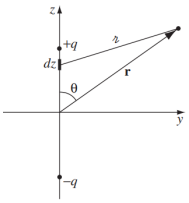
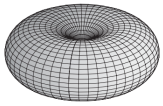
$$\frac{1}{\mathscr{r}_\pm} \cong \frac{1}{r}\left(1 \pm \frac{d}{2r}\cos\theta\right)$$

and:

$$\cos[\omega(t - \mathscr{r}_\pm/c)] \cong \cos\left[\omega(t - r/c) \pm \frac{\omega d}{2c}\cos\theta\right]$$

$$= \cos[\omega(t - r/c)]\cos\left(\frac{\omega d}{2c}\cos\theta\right)$$

$$\mp \sin[\omega(t - r/c)]\sin\left(\frac{\omega d}{2c}\cos\theta\right)$$

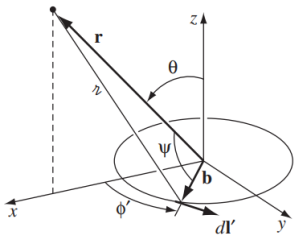
2: d<<c/ω	this corresponds to d<<λ since λ=2πc/ω > under these conditions we have: $$\cos[\omega(t-\imath_\pm/c)] \cong \cos[\omega(t-r/c)] \mp \frac{\omega d}{2c}\cos\theta\sin[\omega(t-r/c)].$$ now putting together the results of the first and second approximation, we have: $$V(r,\theta,t) = \frac{p_0\cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin[\omega(t-r/c)] + \frac{1}{r}\cos[\omega(t-r/c)]\right\}$$ in the static case ω=0, this produces the old formula for a dipole: $$V = \frac{p_0\cos\theta}{4\pi\epsilon_0 r^2}.$$ > this term is not what we are interested in, since static sources don't radiate > we are looking for the fields that survive at large distances from the source
3: r>>c/ω	in this region the potential reduces to: $$V(r,\theta,t) = -\frac{p_0\omega}{4\pi\epsilon_0 c}\left(\frac{\cos\theta}{r}\right)\sin[\omega(t-r/c)].$$
fields **E** and **B** of the oscillating dipole	We can consider the oscillating dipole as two charges going through a wire  > the current flowing through the wire is: $$\mathbf{I}(t) = \frac{dq}{dt}\hat{\mathbf{z}} = -q_0\omega\sin(\omega t)\,\hat{\mathbf{z}}.$$ therefore: $$\mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int_{-d/2}^{d/2}\frac{-q_0\omega\sin[\omega(t-\imath/c)]\,\hat{\mathbf{z}}}{\imath}\,dz.$$ the integration introduces a factor of d > we can, to first order, replace the integrand by its value at the centre: $$\mathbf{A}(r,\theta,t) = -\frac{\mu_0 p_0\omega}{4\pi r}\sin[\omega(t-r/c)]\,\hat{\mathbf{z}}.$$ now, from the potentials: $$\begin{aligned}\nabla V &= \frac{\partial V}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial V}{\partial\theta}\hat{\boldsymbol{\theta}}\\ &= -\frac{p_0\omega}{4\pi\epsilon_0 c}\left\{\cos\theta\left(-\frac{1}{r^2}\sin[\omega(t-r/c)] - \frac{\omega}{rc}\cos[\omega(t-r/c)]\right)\hat{\mathbf{r}}\right.\\ &\qquad\left. - \frac{\sin\theta}{r^2}\sin[\omega(t-r/c)]\,\hat{\boldsymbol{\theta}}\right\}\\ &\cong \frac{p_0\omega^2}{4\pi\epsilon_0 c^2}\left(\frac{\cos\theta}{r}\right)\cos[\omega(t-r/c)]\,\hat{\mathbf{r}}.\end{aligned}$$ and: $$\frac{\partial\mathbf{A}}{\partial t} = -\frac{\mu_0 p_0\omega^2}{4\pi r}\cos[\omega(t-r/c)](\cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}),$$ and: $$\begin{aligned}\nabla\times\mathbf{A} &= \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\hat{\boldsymbol{\phi}}\\ &= -\frac{\mu_0 p_0\omega}{4\pi r}\left\{\frac{\omega}{c}\sin\theta\cos[\omega(t-r/c)] + \frac{\sin\theta}{r}\sin[\omega(t-r/c)]\right\}\hat{\boldsymbol{\phi}}.\end{aligned}$$ we find the **E** and **B** fields: $$\mathbf{E} = -\nabla V - \frac{\partial\mathbf{A}}{\partial t} = -\frac{\mu_0 p_0\omega^2}{4\pi}\left(\frac{\sin\theta}{r}\right)\cos[\omega(t-r/c)]\,\hat{\boldsymbol{\theta}}.$$ $$\mathbf{B} = \nabla\times\mathbf{A} = -\frac{\mu_0 p_0\omega^2}{4\pi c}\left(\frac{\sin\theta}{r}\right)\cos[\omega(t-r/c)]\,\hat{\boldsymbol{\phi}}.$$ >> fields are in phase, mutually perpendicular and transverse with $E_0/B_0=c$
radiated energy	the Poynting vector: $$\mathbf{S}(\mathbf{r},t) = \frac{1}{\mu_0}(\mathbf{E}\times\mathbf{B}) = \frac{\mu_0}{c}\left\{\frac{p_0\omega^2}{4\pi}\left(\frac{\sin\theta}{r}\right)\cos[\omega(t-r/c)]\right\}^2\hat{\mathbf{r}}.$$  then the average intensity over time is: $$\langle\mathbf{S}\rangle = \left(\frac{\mu_0 p_0^2\omega^4}{32\pi^2 c}\right)\frac{\sin^2\theta}{r^2}\hat{\mathbf{r}}.$$ > notice that there is no radiation along the axis of the pole, radiation is in a donut The total power radiated is then: $\langle P\rangle = \int\langle\mathbf{S}\rangle\cdot d\mathbf{a} = \frac{\mu_0 p_0^2\omega^4}{32\pi^2 c}\int\frac{\sin^2\theta}{r^2}r^2\sin\theta\,d\theta\,d\phi = \frac{\mu_0 p_0^2\omega^4}{12\pi c}.$

11.1.3 magnetic dipole radiation

oscillating magnetic dipole

consider a wire loop of radius b around which we drive an alternating current:

$$I(t) = I_0 \cos(\omega t).$$



then, for a maximum value $m_0=\pi b^2 I_0$:

$$\mathbf{m}(t) = \pi b^2 I(t)\,\hat{\mathbf{z}} = m_0 \cos(\omega t)\,\hat{\mathbf{z}},$$

the retarded vector potential is:

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{I_0 \cos[\omega(t - \imath/c)]}{\imath}\, d\mathbf{l}'.$$

for a point **r** directly above the x-axis, **A** must aim in the y direction:

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 I_0 b}{4\pi}\,\hat{\mathbf{y}} \int_0^{2\pi} \frac{\cos[\omega(t - \imath/c)]}{\imath} \cos\phi'\, d\phi'$$

furthermore we know the law of cosines:

$$\imath = \sqrt{r^2 + b^2 - 2rb\sin\theta\cos\phi'}.$$

and the approximations:

1: b<<r
2: b<<c/ω
3: r>>c/ω

1: b<<r

to the first order:

$$\imath \cong r\left(1 - \frac{b}{r}\sin\theta\cos\phi'\right) \qquad <=> \qquad \frac{1}{\imath} \cong \frac{1}{r}\left(1 + \frac{b}{r}\sin\theta\cos\phi'\right)$$

and:

$$\begin{aligned}\cos[\omega(t - \imath/c)] &\cong \cos\left[\omega(t - r/c) + \frac{\omega b}{c}\sin\theta\cos\phi'\right] \\ &= \cos[\omega(t - r/c)]\cos\left(\frac{\omega b}{c}\sin\theta\cos\phi'\right) \\ &\quad - \sin[\omega(t - r/c)]\sin\left(\frac{\omega b}{c}\sin\theta\cos\phi'\right)\end{aligned}$$

2: b<<c/ω

we have:

$$\cos[\omega(t - \imath/c)] \cong \cos[\omega(t - r/c)] - \frac{\omega b}{c}\sin\theta\cos\phi'\sin[\omega(t - r/c)].$$

with this the vector potential becomes:

$$\begin{aligned}\mathbf{A}(\mathbf{r}, t) \cong \frac{\mu_0 I_0 b}{4\pi r}\,\hat{\mathbf{y}} \int_0^{2\pi} \Big\{ &\cos[\omega(t - r/c)] + b\sin\theta\cos\phi' \\ &\times \left(\frac{1}{r}\cos[\omega(t - r/c)] - \frac{\omega}{c}\sin[\omega(t - r/c)]\right)\Big\}\cos\phi'\, d\phi'.\end{aligned}$$

the first term integrates to zero:

$$\int_0^{2\pi} \cos\phi'\, d\phi' = 0.$$

the second term to π:

$$\int_0^{2\pi} \cos^2\phi'\, d\phi' = \pi.$$

thus for a oscillating perfect magnetic dipole:

$$\mathbf{A}(r, \theta, t) = \frac{\mu_0 m_0}{4\pi}\left(\frac{\sin\theta}{r}\right)\left\{\frac{1}{r}\cos[\omega(t - r/c)] - \frac{\omega}{c}\sin[\omega(t - r/c)]\right\}\hat{\boldsymbol{\phi}}.$$

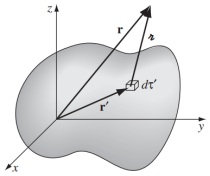
in the static limit ω=0, we recover the formula we already know:

$$\mathbf{A}(r, \theta) = \frac{\mu_0}{4\pi}\frac{m_0 \sin\theta}{r^2}\,\hat{\boldsymbol{\phi}}.$$

3: r>>c/ω

we find:

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 m_0 \omega}{4\pi c}\left(\frac{\sin\theta}{r}\right)\sin[\omega(t - r/c)]\,\hat{\boldsymbol{\phi}}.$$

E and **B** of the dipole	from **A** we obtain at large r: $$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2}{4\pi c}\left(\frac{\sin\theta}{r}\right)\cos[\omega(t - r/c)]\,\hat{\boldsymbol{\phi}},$$ $$\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\mu_0 m_0 \omega^2}{4\pi c^2}\left(\frac{\sin\theta}{r}\right)\cos[\omega(t - r/c)]\,\hat{\boldsymbol{\theta}}.$$ >> these fields are in phase, mutually perpendicular and transverse in direction of propagation > the ratio of amplitudes $E_0/B_0 = c$
intensity and total radiation of the dipole	the energy flux is: $$\mathbf{S}(\mathbf{r},t) = \frac{1}{\mu_0}(\mathbf{E}\times\mathbf{B}) = \frac{\mu_0}{c}\left\{\frac{m_0\omega^2}{4\pi c}\left(\frac{\sin\theta}{r}\right)\cos[\omega(t-r/c)]\right\}^2\hat{\mathbf{r}},$$ then the intensity is: $$\langle \mathbf{S}\rangle = \left(\frac{\mu_0 m_0^2\omega^4}{32\pi^2 c^3}\right)\frac{\sin^2\theta}{r^2}\hat{\mathbf{r}},$$ and the total radiated power: $$\langle P\rangle = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}.$$
electric radiation compared to magnetic radiation	if we compare the total radiated power: $$\frac{P_{\text{magnetic}}}{P_{\text{electric}}} = \left(\frac{m_0}{p_0 c}\right)^2,$$ or thus: $$\frac{P_{\text{magnetic}}}{P_{\text{electric}}} = \left(\frac{\omega b}{c}\right)^2.$$ where ωb/c is very small > electric dipole radiation dominates the magnetic dipole radiation
11.1.4 radiation form an arbitrary source	
arbitrary source of em	Consider an arbitrary configuration of charge and current in some finite volume near the origin > the retarded scalar potential is: $$V(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}', t - \imath/c)}{\imath}\,d\tau',$$ where we know: $$\imath = \sqrt{r^2 + r'^2 - 2\mathbf{r}\cdot\mathbf{r}'}.$$ where we can make approximations: 1: r'<<r 2: $r' \ll \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert}, \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert^{1/2}}, \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert^{1/3}}, \ldots$ 3: discard $1/r^2$ terms in **E** and **B** 
1: r'<<r	with this approximation we have: $\imath \cong r\left(1 - \frac{\mathbf{r}\cdot\mathbf{r}'}{r^2}\right)$ <=> $\frac{1}{\imath} \cong \frac{1}{r}\left(1 + \frac{\mathbf{r}\cdot\mathbf{r}'}{r^2}\right)$ and: $$\rho(\mathbf{r}', t - \imath/c) \cong \rho\left(\mathbf{r}', t - \frac{r}{c} + \frac{\hat{\mathbf{r}}\cdot\mathbf{r}'}{c}\right)$$ expand ρ as a Taylor series in t about the retarded time at the origin t_0=t-r/c: $$\rho(\mathbf{r}', t - \imath/c) \cong \rho(\mathbf{r}', t_0) + \dot{\rho}(\mathbf{r}', t_0)\left(\frac{\hat{\mathbf{r}}\cdot\mathbf{r}'}{c}\right) + \ldots$$
2: $r' \ll \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert}, \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert^{1/2}}, \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert^{1/3}}, \ldots$	we can drop the second order terms of the Taylor expansion, and thus: $$\rho(\mathbf{r}', t - \imath/c) \cong \rho(\mathbf{r}', t_0) + \dot{\rho}(\mathbf{r}', t_0)\left(\frac{\hat{\mathbf{r}}\cdot\mathbf{r}'}{c}\right)$$ putting all of this in the eq for V, we find: $$V(\mathbf{r},t) \cong \frac{1}{4\pi\epsilon_0 r}\left[\int \rho(\mathbf{r}', t_0)\,d\tau' + \frac{\hat{\mathbf{r}}}{r}\cdot\int \mathbf{r}'\rho(\mathbf{r}', t_0)\,d\tau' + \frac{\hat{\mathbf{r}}}{c}\cdot\frac{d}{dt}\int \mathbf{r}'\rho(\mathbf{r}', t_0)\,d\tau'\right]$$ the first integral is the total charge Q at time t_0 > charge is conserved, thus Q is time-independent > furthermore, the other two integrals represent the electric dipole moment at t_0 > we then find: $$V(\mathbf{r},t) \cong \frac{1}{4\pi\epsilon_0}\left[\frac{Q}{r} + \frac{\hat{\mathbf{r}}\cdot\mathbf{p}(t_0)}{r^2} + \frac{\hat{\mathbf{r}}\cdot\dot{\mathbf{p}}(t_0)}{rc}\right]$$

2: $r' \ll \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert}, \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert^{1/2}}, \frac{c}{\lvert\ddot{\rho}/\dot{\rho}\rvert^{1/3}}, \ldots$	the vector potential is: $$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t - \imath/c)}{\imath} d\tau'.$$ to the first order in r' we can replace $\imath$ by r: $$\mathbf{A}(\mathbf{r}, t) \cong \frac{\mu_0}{4\pi r} \int \mathbf{J}(\mathbf{r}', t_0)\, d\tau'.$$ <=> $$\mathbf{A}(\mathbf{r}, t) \cong \frac{\mu_0}{4\pi} \frac{\dot{\mathbf{p}}(t_0)}{r}.$$
3: discard $1/r^2$ terms in **E** & **B**	from t_0=t-r/c, we find: $\nabla t_0 = -\frac{1}{c}\nabla r = -\frac{1}{c}\hat{\mathbf{r}}$, hence: $$\nabla V \cong \nabla \left[\frac{1}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}} \cdot \dot{\mathbf{p}}(t_0)}{rc}\right] \cong \frac{1}{4\pi\epsilon_0} \left[\frac{\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}(t_0)}{rc}\right] \nabla t_0 = -\frac{1}{4\pi\epsilon_0 c^2} \frac{[\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}(t_0)]}{r} \hat{\mathbf{r}}.$$ and: $$\nabla \times \mathbf{A} \cong \frac{\mu_0}{4\pi r}[\nabla \times \dot{\mathbf{p}}(t_0)] = \frac{\mu_0}{4\pi r}[(\nabla t_0) \times \ddot{\mathbf{p}}(t_0)] = -\frac{\mu_0}{4\pi rc}[\hat{\mathbf{r}} \times \ddot{\mathbf{p}}(t_0)],$$ while: $$\frac{\partial \mathbf{A}}{\partial t} \cong \frac{\mu_0}{4\pi} \frac{\ddot{\mathbf{p}}(t_0)}{r}.$$ we then find: $$\mathbf{E}(\mathbf{r}, t) \cong \frac{\mu_0}{4\pi r}[(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}})\hat{\mathbf{r}} - \ddot{\mathbf{p}}] = \frac{\mu_0}{4\pi r}[\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \ddot{\mathbf{p}})],$$ $$\mathbf{B}(\mathbf{r}, t) \cong -\frac{\mu_0}{4\pi rc}[\hat{\mathbf{r}} \times \ddot{\mathbf{p}}].$$ where $\ddot{\mathbf{p}}$ is evaluated at time $t_0 = t - r/c$, and
Poynting vector and radiated power	The Poynting vector is: $$\mathbf{S}(\mathbf{r}, t) \cong \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}) = \frac{\mu_0}{16\pi^2 c}[\ddot{p}(t_0)]^2 \left(\frac{\sin^2\theta}{r^2}\right) \hat{\mathbf{r}},$$ the power passing through a giant spherical surface at radius r is: $$P(r, t) = \oint \mathbf{S}(\mathbf{r}, t) \cdot d\mathbf{a} = \frac{\mu_0}{6\pi c}\left[\ddot{p}\left(t - \frac{r}{c}\right)\right]^2,$$ the total radiated power is: $$P_{\text{rad}}(t_0) \cong \frac{\mu_0}{6\pi c}\left[\ddot{p}(t_0)\right]^2.$$